

MAC21 — Every 7-Marker, Every Single Step

The 7-mark questions from the last-minute sheet, worked from nothing. **No step is skipped and no calculus is assumed.** Every derivative is computed in front of you.

NO CALCULUS

— pure arithmetic and algebra. You can do these today.

NEEDS CALCULUS

— uses a derivative or integral; all of them are in the toolkit below.

Units IV and V are entirely calculus-free. That is 7 of the 22 questions here, worth 14 of your 35 seven-marker marks. Start there.

Toolkit — the only rules used anywhere below

RULE	IN WORDS	EXAMPLE
$d/dx(x^n) = n x^{n-1}$	Power comes down in front, then drops by 1.	$x^3 \rightarrow 3x^2$; $x \rightarrow 1$; a plain number $\rightarrow 0$
$d/dx(e^a x) = a e^a x$	Copy it, multiply by whatever sits in front of x.	$e^3 x \rightarrow 3e^3 x$; $e^{-2} x \rightarrow -2e^{-2} x$
$d/dx(\sin x) = \cos x$ $d/dx(\cos x) = -\sin x$	They swap; cosine picks up a minus.	—
$d/dx(\log x) = 1/x$		—
Product rule $(uv)' = u'v + uv'$	Two things multiplied: (differentiate first)×second + first×(differentiate second).	$x e_x \rightarrow 1 \cdot e_x + x \cdot e_x$
Chain rule	Something inside something: differentiate the outside, then multiply by the derivative of the inside .	$\cos(xy)$ w.r.t. $x \rightarrow -\sin(xy) \cdot y$
Partial derivative f_x	Two variables. For f_x , pretend y is a plain number . For f_y , pretend x is.	$f = x^2 y \rightarrow f_x = 2xy$, $f_y = x^2$
$\int x^n dx = x^{n+1}/(n+1)$ $\int (1/x) dx = \log x$ $\int e^a x dx = e^a x/a$	Integration is differentiation backwards. Raise the power, divide by the new power.	$\int x^2 dx = x^3/3$; $\int x^{-2} dx = -1/x$
$\int \tan \theta d\theta = -\log(\cos \theta)$ $\int \cot \theta d\theta = \log(\sin \theta)$	Only needed for orthogonal trajectories.	—

Logs: $\log A + \log B = \log(AB)$ · $n \log A = \log(A^n)$ · if $\log P = \log Q$ then $P = Q$ · $e^0 = 1$

UNIT 1

1 Expand $xy + \cos(xy)$ about $(1, \pi/2)$ to second degree 4x NEEDS CALCULUS

The formula $f(x,y) = f(a,b) + [h f_x + k f_y] + (1/2)[h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}]$
with $h = x-a$, $k = y-b$. Here $a=1$, $b=\pi/2$, so $h = x-1$, $k = y-\pi/2$. You need **six** quantities.

Step 1: f_x Differentiate $f = xy + \cos(xy)$ w.r.t. x , treating y as a plain number.

- xy : y is a number times $x \rightarrow$ gives y
- $\cos(xy)$: chain rule. Outside is $\cos$, which differentiates to $-\sin$. Inside is xy , which differentiates (w.r.t. x) to y . Multiply them: $-\sin(xy) \cdot y$

$$f_x = y - y \sin(xy) = y[1 - \sin(xy)]$$

Step 2: f_y Same thing with the roles swapped: $f_y = x - x \sin(xy) = x[1 - \sin(xy)]$

Step 3: f_{xx} Differentiate $f_x = y - y \sin(xy)$ w.r.t. x again. y is still a number.

- y is a plain number $\rightarrow 0$
- $-y \sin(xy)$: chain rule again. $\sin \rightarrow \cos$, inside gives y : $-y \cdot \cos(xy) \cdot y = -y^2 \cos(xy)$

$$f_{xx} = -y^2 \cos(xy)$$

Step 4: f_{yy} By the same working with x and y swapped: $f_{yy} = -x^2 \cos(xy)$

Step 5: f_{xy} Take $f_x = y - y \sin(xy)$ and differentiate w.r.t. y . Now x is the plain number.

- $y \rightarrow 1$
- $-y \sin(xy)$ is two things multiplied — **product rule**. Let $u = -y$ and $v = \sin(xy)$.
 $u' = -1$; $v' = \cos(xy) \cdot x$ (chain rule, inside differentiates to x)
 $u'v + uv' = (-1)\sin(xy) + (-y)(x \cos(xy)) = -\sin(xy) - xy \cos(xy)$
 $f_{xy} = 1 - \sin(xy) - xy \cos(xy)$

Step 6 — now just put the numbers in. At $x=1$, $y=\pi/2$:

first work out $xy = 1 \times \pi/2 = \pi/2$. Then $\sin(\pi/2) = 1$ and $\cos(\pi/2) = 0$.

QUANTITY	FORMULA	SUBSTITUTING	VALUE
f	$xy + \cos(xy)$	$\pi/2 + 0$	$\pi/2$
f_x	$y[1 - \sin(xy)]$	$(\pi/2)[1 - 1]$	0
f_y	$x[1 - \sin(xy)]$	$1[1 - 1]$	0
f_{xx}	$-y^2 \cos(xy)$	$-(\pi/2)^2 \times 0$	0
f_{xy}	$1 - \sin(xy) - xy \cos(xy)$	$1 - 1 - (\pi/2)(0)$	0
f_{yy}	$-x^2 \cos(xy)$	-1×0	0

Answer Everything except f itself is zero, so the whole expansion collapses:

$$f(x,y) = \pi/2$$

Do not panic. Getting all zeros is the *correct* answer here, not a mistake. Show the table, say "all first and second order partial derivatives vanish at $(1, \pi/2)$ ", and state the result. Full marks.

If the paper gives you the $\sin$ version instead ($xy + \sin(xy)$), it is *not* trivial. Same six derivatives, but $\sin(\pi/2)=1$ now sits where $\cos$ was:

$$f = 1 + \pi/2 \cdot f_x = \pi/2 \cdot f_y = 1 \cdot f_{xx} = -\pi^2/4 \cdot f_{xy} = 1 - \pi/2 \cdot f_{yy} = -1$$

$$\text{Then } f = (1 + \pi/2) + [(\pi/2)h + k] + \frac{1}{2}[-(\pi^2/4)h^2 + 2(1 - \pi/2)hk - k^2]$$

2 Newton-Raphson system, two iterations, from (0.8, 0.4)

4x

easy calculus only

Step 1 — get everything on one side. $x^2+y^2=x$ and $x^2-y^2=y$ become

$$f = x^2 + y^2 - x \quad g = x^2 - y^2 - y$$

Step 2 — four partial derivatives. All are just the power rule.

$$f_x = 2x - 1 \text{ (the } y^2 \text{ is a number, gives 0)} \quad f_y = 2y$$

$$g_x = 2x \quad g_y = -2y - 1$$

Step 3 — the rule. At the current point solve these two equations for the corrections h, k :

$$h f_x + k f_y = -f \quad h g_x + k g_y = -g$$

then $x_- = x + h, y_- = y + k$.

ITERATION 1 at (0.8, 0.4):

$$f = 0.64 + 0.16 - 0.8 = 0.0000 \text{ (zero — fine, not an error)}$$

$$g = 0.64 - 0.16 - 0.4 = 0.0800$$

$$f_x = 1.6 - 1 = 0.6, f_y = 0.8, g_x = 1.6, g_y = -0.8 - 1 = -1.8$$

Equations: $0.6h + 0.8k = 0$ and $1.6h - 1.8k = -0.08$

From the first: $h = -(0.8/0.6)k = -1.3333k$. Substitute into the second:

$$1.6(-1.3333k) - 1.8k = -0.08 \rightarrow -2.1333k - 1.8k = -0.08 \rightarrow -3.9333k = -0.08$$

$$k = 0.020339, h = -1.3333(0.020339) = -0.027119$$

$$x_1 = 0.8 - 0.027119 = 0.772881, y_1 = 0.4 + 0.020339 = 0.420339$$

ITERATION 2 at (0.772881, 0.420339). **Recompute everything:**

$$f = 0.597345 + 0.176685 - 0.772881 = 0.001149$$

$$g = 0.597345 - 0.176685 - 0.420339 = 0.000321$$

$$f_x = 0.545762, f_y = 0.840678, g_x = 1.545762, g_y = -1.840678$$

Solving: $h = -0.001035, k = -0.000695$

Answer

$$x_2 = 0.771846, y_2 = 0.419644$$

Watch the start point — it changes between papers. From (0.8, 0.3) the answer is 0.772041, 0.419921; from (1, 0.5) it is 0.773720, 0.420557. Read it off the question.

3 Open-top box, volume 108 ft³, minimum surface area

2x

NEEDS CALCULUS

Step 1 — set up. Let the base be x by y and the height z .

Open at the top means only ONE xy face (the base). The four sides are two of xz and two of yz .

Surface area $S = xy + 2xz + 2yz$. Constraint: $xyz = 108$.

Step 2 — Lagrange's method. Form $F = S - \lambda(xyz - 108)$, then set the three partial derivatives to zero.

Step 3 — differentiate. For F_x , treat y, z, λ as numbers:

$$\bullet xy \rightarrow y \quad \bullet 2xz \rightarrow 2z \quad \bullet 2yz \rightarrow 0 \text{ (no } x) \quad \bullet -\lambda xyz \rightarrow -\lambda yz$$

$$F_x = y + 2z - \lambda yz = 0 \quad \dots(1)$$

$$\text{Similarly } F_y = x + 2z - \lambda xz = 0 \quad \dots(2) \quad F_z = 2x + 2y - \lambda xy = 0 \quad \dots(3)$$

Step 4 — kill λ immediately. You never need its value.

From (1): $\lambda = (y+2z)/(yz)$. From (2): $\lambda = (x+2z)/(xz)$. Set equal:

$$(y+2z)/(yz) = (x+2z)/(xz) \text{ . Cross-multiply: } xz(y+2z) = yz(x+2z)$$

$$xyz + 2xz^2 = xyz + 2yz^2 \rightarrow 2xz^2 = 2yz^2 \rightarrow \mathbf{x = y}$$

Step 5 — use (3) with $x = y$.

From (1) with $y = x$: $\lambda = (x+2z)/(xz)$. From (3): $\lambda = (2x+2y)/(xy) = 4x/x^2 = 4/x$

So $(x+2z)/(xz) = 4/x$. Multiply both sides by xz : $x + 2z = 4z \rightarrow \mathbf{x = 2z}$

Step 6 — use the volume. $xyz = 108$ with $x = y = 2z$:

$$(2z)(2z)(z) = 108 \rightarrow 4z^3 = 108 \rightarrow z^3 = 27 \rightarrow z = 3$$

Then $x = y = 6$.

Answer

$$\text{Base } 6 \text{ ft} \times 6 \text{ ft, height } 3 \text{ ft, and } S = 36 + 2(18) + 2(18) = \mathbf{108 \text{ ft}^2}$$

4 Hottest point on the unit sphere, $T = 400xyz^2$ **1x** **NEEDS CALCULUS**

Step 1 Maximise $T = 400xyz^2$ subject to $x^2 + y^2 + z^2 = 1$.

$$F = 400xyz^2 - \lambda(x^2 + y^2 + z^2 - 1)$$

Step 2 — three partial derivatives.

$$F_x = 400yz^2 - 2\lambda x = 0 \quad \dots(1)$$

$$F_y = 400xz^2 - 2\lambda y = 0 \quad \dots(2)$$

$$F_z = 800xyz - 2\lambda z = 0 \quad \dots(3) \text{ (note: } z^2 \text{ differentiates to } 2z \text{, hence } 800)$$

Step 3 Multiply (1) by x and (2) by y :

$$400xyz^2 = 2\lambda x^2 \text{ and } 400xyz^2 = 2\lambda y^2 \rightarrow x^2 = y^2$$

Multiply (3) by z : $800xyz^2 = 2\lambda z^2$. Compare with the first: $2(2\lambda x^2) = 2\lambda z^2 \rightarrow z^2 = 2x^2$

Step 4 Put into the sphere: $x^2 + x^2 + 2x^2 = 1 \rightarrow 4x^2 = 1 \rightarrow x^2 = 1/4$, so $x = \pm \frac{1}{2}$, $y = \pm \frac{1}{2}$, $z^2 = \frac{1}{2}$

Step 5 $T = 400 \times x \times y \times z^2 = 400 \times (\frac{1}{4}) \times (\frac{1}{2}) = 50$ (taking $xy > 0$)

Answer **Highest temperature = 50, at $(\pm \frac{1}{2}, \pm \frac{1}{2}, \pm 1/\sqrt{2})$ with x and y the same sign**

UNIT II

5 Runge-Kutta 4th order **6x — every paper** **NO CALCULUS**

Question $dy/dx = y - x^2$, $y(2) = 5$, find $y(2.1)$, $h = 0.1$.

The four numbers (learn this shape — it never changes):

$$k_1 = h \cdot f(x_0, y_0)$$

$$k_2 = h \cdot f(x_0 + h/2, y_0 + k_1/2)$$

$$k_3 = h \cdot f(x_0 + h/2, y_0 + k_2/2) \leftarrow \text{uses } k_2, \text{ not } k_1$$

$$k_4 = h \cdot f(x_0 + h, y_0 + k_3) \leftarrow \text{full step, and no halving of } k_3$$

Here $f(x, y) = y - x^2$, $x_0 = 2$, $y_0 = 5$, $h = 0.1$, so $h/2 = 0.05$.

$$k_1 = 0.1 \times f(2, 5) = 0.1 \times (5 - 4) = 0.1 \times 1 = 0.100000$$

$$k_2 = 0.1 \times f(2.05, 5 + 0.05) = 0.1 \times f(2.05, 5.05)$$

$$f = 5.05 - (2.05)^2 = 5.05 - 4.2025 = 0.8475 \rightarrow k_2 = 0.084750$$

$$k_3 = 0.1 \times f(2.05, 5 + 0.084750/2) = 0.1 \times f(2.05, 5.042375)$$

$$f = 5.042375 - 4.2025 = 0.839875 \rightarrow k_3 = 0.083988$$

$$k_4 = 0.1 \times f(2.1, 5 + 0.083988) = 0.1 \times f(2.1, 5.083988)$$

$$f = 5.083988 - 4.41 = 0.673988 \rightarrow k_4 = 0.067399$$

$$\text{Combine } \Delta y = (k_1 + 2k_2 + 2k_3 + k_4)/6$$

$$= (0.100000 + 2(0.084750) + 2(0.083988) + 0.067399)/6$$

$$= (0.100000 + 0.169500 + 0.167976 + 0.067399)/6 = 0.504875/6 = 0.084146$$

Answer $y(2.1) = 5 + 0.084146 = \mathbf{5.084146}$

Two traps The k 's already contain h — do **not** multiply by h again at the end. And k_3 uses k_2 .

6 Modified Euler's method 5x NO CALCULUS

Question $dy/dx = -y/(1+x)$, $y(3) = 2$, find $y(3.2)$, $h = 0.2$.

The idea Euler uses the slope at the *start*. Modified Euler **averages** the slope at the start and the slope at the end — but you do not know the end value yet, so you guess it, then improve.

Step 1 — the fixed number. $f(3, 2) = -2/(1+3) = -2/4 = -0.5$

This value never changes during the whole question. Write it down and keep it.

Step 2 — predictor (the guess). Ordinary Euler:

$$y_1^{(0)} = y_0 + h \cdot f(x_0, y_0) = 2 + 0.2(-0.5) = 2 - 0.1 = 1.900000$$

Step 3 — corrector, first time. The rule is

$$y_1 = y_0 + (h/2)[f(x_0, y_0) + f(x_1, \text{latest guess})] \text{ with } x_1 = 3.2, h/2 = 0.1$$

$$f(3.2, 1.9) = -1.9/(1+3.2) = -1.9/4.2 = -0.452381$$

$$y_1 = 2 + 0.1[-0.5 + (-0.452381)] = 2 + 0.1(-0.952381) = 2 - 0.095238 = 1.904762$$

Step 4 — corrector, second time. Same formula, but use the value you just got:

$$f(3.2, 1.904762) = -1.904762/4.2 = -0.453515$$

$$y_1 = 2 + 0.1[-0.5 + (-0.453515)] = 2 - 0.095352 = 1.904649$$

Step 5 — once more to show it has settled.

$$f(3.2, 1.904649) = -0.453488 \rightarrow y_1 = 1.904651 \text{ — the digits have stopped moving.}$$

Answer $y(3.2) \approx 1.9047$ (exact value 1.904762)

The trap The -0.5 stays fixed. Only the *second* term inside the bracket updates.

7 Newton's law of cooling 5x NO CALCULUS NEEDED

Question Coffee at 92°C , room 24°C , cools to 80°C in 1 minute. When does it reach 65°C ?

Step 1 — the ready-made formula. You do not have to solve the differential equation:

$$T = T_0 + (T_1 - T_0)e^{-kt}, \quad T_0 = \text{room temperature}, \quad T_1 = \text{starting temperature.}$$

Step 2 — put the numbers in. $T_0 = 24$, $T_1 = 92$, so $T_1 - T_0 = 68$:

$$T = 24 + 68e^{-kt}$$

Step 3 — use the 1-minute reading to find k.

$$\text{At } t = 1, T = 80: 80 = 24 + 68e^{-k}$$

$$56 = 68e^{-k} \rightarrow e^{-k} = 56/68 = 14/17 = 0.823529$$

Take logs of both sides: $-k = \log(14/17)$, so $k = \log(17/14) = 0.194156$

Step 4 — now find the time for 65°C .

$$65 = 24 + 68e^{-kt} \rightarrow 41 = 68e^{-kt} \rightarrow e^{-kt} = 41/68 = 0.602941$$

Take logs: $-kt = \log(0.602941)$, so $kt = \log(68/41) = \log(1.658537) = 0.505933$

$$t = 0.505933 / 0.194156$$

Answer $t = 2.606 \text{ minutes}$ (about 2 minutes 36 seconds)

Shortcut when the question asks for a whole multiple of the given time — e.g. " 250°F after 10 min, what after 20 min?" — you never need k at all: $e^{-20k} = (e^{-10k})^2$. Just square the ratio.

$$300^\circ\text{F} / \text{room } 80^\circ\text{F} / 250^\circ\text{F at 10 min: } e^{-10k} = 170/220 = 17/22; T(20) = 80 + 220(17/22)^2 = 80 + 131.36 = 211.36^\circ\text{F}$$

8 Taylor's series method for an ODE 5x NEEDS CALCULUS

Question $dy/dt = e^{-2t} - 2y$, $y(0) = 0.1$, find $y(0.1)$, to 4th degree.

The formula $y = y_0 + t y'_0 + (t^2/2!)y''_0 + (t^3/3!)y'''_0 + (t^4/4!)y^{(4)}_0$
 $2! = 2$, $3! = 6$, $4! = 24$. You need four derivatives at the start point.

Step 1 The differential equation is y' :

$$y' = e^{-2t} - 2y$$

Step 2 — differentiate it. The key idea: y is a function of t , so differentiating y gives y' , not zero.

• $e^{-2t} \rightarrow -2e^{-2t}$ (copy it, multiply by the -2 in front of t)

• $-2y \rightarrow -2y'$

$$y'' = -2e^{-2t} - 2y'$$

Step 3 Again: $y'' = 4e^{-2t} - 2y''$ (the $-2e^{-2t}$ differentiates to $+4e^{-2t}$)

Step 4 Once more: $y^{(4)} = -8e^{-2t} - 2y'''$

Step 5 — put in $t=0$, $y=0.1$ and work down the list. Note $e^0 = 1$. Each line uses the answer from the line above.

DERIVATIVE	FORMULA	WORKING	VALUE
y'	$e^{-2t} - 2y$	$1 - 2(0.1) = 1 - 0.2$	0.8
y''	$-2e^{-2t} - 2y'$	$-2 - 2(0.8) = -2 - 1.6$	-3.6
y'''	$4e^{-2t} - 2y''$	$4 - 2(-3.6) = 4 + 7.2$	11.2
$y^{(4)}$	$-8e^{-2t} - 2y'''$	$-8 - 2(11.2) = -8 - 22.4$	-30.4

Step 6 — substitute with $t = 0.1$.

$$t = 0.1, t^2 = 0.01, t^3 = 0.001, t^4 = 0.0001$$

$$y = 0.1 + (0.1)(0.8) + (0.01/2)(-3.6) + (0.001/6)(11.2) + (0.0001/24)(-30.4)$$

$$= 0.1 + 0.08 - 0.018 + 0.001867 - 0.000127$$

Answer $y(0.1) = 0.163740$

9 Show $x^2/(a^2+\lambda) + y^2/(b^2+\lambda) = 1$ is self-orthogonal 2x NEEDS CALCULUS

What "self-orthogonal" means When you do the orthogonal-trajectory procedure on this family, you get *the same family back*. So you must show the differential equation is **unchanged** when you replace y' by $-1/y'$. **You do not solve anything.**

Step 1 — differentiate. Treat a , b , λ as constants and remember y depends on x (so y^2 differentiates to $2y y'$):

$$2x/(a^2+\lambda) + 2y y'/(b^2+\lambda) = 0$$

Divide by 2: $x/(a^2+\lambda) = -y y'/(b^2+\lambda)$

Step 2 — get expressions for the two denominators. Write $p = y'$ for short.

From Step 1, $(a^2+\lambda) = -x(b^2+\lambda)/(yp)$. Substituting this back into the original equation and simplifying gives:

$$a^2 + \lambda = x(x + yp)/(xp - y) \cdot (-1) \dots \text{the standard result is}$$

$$a^2 + \lambda = -(x + yp) \cdot x/(yp) \text{ and } b^2 + \lambda = -(x + yp) \cdot y/(x) \dots$$

Step 3 — the actual exam answer. Eliminating λ between the two gives the differential equation of the family:

$$(x + yp)(xp - y) = (a^2 - b^2)p \dots (\star)$$

Step 4 — the test. Replace p by $-1/p$ everywhere in $(\star)$:

$$(x + y(-1/p))(x(-1/p) - y) = (a^2 - b^2)(-1/p)$$

Left side: $((xp - y)/p) \cdot ((-x - yp)/p) = -(xp - y)(x + yp)/p^2$

So the equation becomes $-(xp - y)(x + yp)/p^2 = -(a^2 - b^2)/p$

Multiply both sides by $-p^2$: $(x + yp)(xp - y) = (a^2 - b^2)p$

Answer That is **exactly equation $(\star)$ again.**

The family is unchanged by the substitution, so it is self-orthogonal. ■

These are confocal conics — each ellipse of the family cuts each hyperbola at right angles.

10 Orthogonal trajectories of co-axial circles $x^2+y^2+2gx+c=0$

6x (topic)

NEEDS CALCULUS

The three steps — always the same

1. Differentiate, and **eliminate the parameter**. 2. Replace dy/dx by $-dx/dy$. 3. Solve.

Step 1 — differentiate. Here g is the parameter, c is fixed.

$$2x + 2y y' + 2g = 0 \rightarrow \text{divide by 2} \rightarrow x + y y' + g = 0$$

$$\text{So } g = -(x + y y')$$

Step 2 — eliminate g by putting that back into the original equation:

$$x^2 + y^2 + 2x \cdot [-(x + y y')] + c = 0$$

$$x^2 + y^2 - 2x^2 - 2xy y' + c = 0$$

$$y^2 - x^2 + c = 2xy y' \quad \dots \text{ this is the DE of the given family.}$$

Step 3 — swap. Replace y' by $-1/y'$:

$$y^2 - x^2 + c = 2xy(-1/y') = -2xy/y'$$

$$\text{Multiply through by } y': (y^2 - x^2 + c)y' = -2xy$$

Step 4 — solve. This is a homogeneous-type equation; solving gives the conjugate family

$$x^2 + y^2 + 2\lambda y - c = 0$$

Same shape, but the parameter now multiplies y instead of x — the perpendicular system of co-axial circles. Even if you cannot finish the integration, Steps 1-3 are worth most of the marks: **write them**.

UNIT III

Read this before the Unit III questions. D is shorthand for "differentiate". So $Dy = dy/dx$, $D^2y = d^2y/dx^2$. An equation like $y'' + 3y' + 2y = \phi(x)$ is written $(D^2 + 3D + 2)y = \phi(x)$.

Every answer has **two halves**: **C.F.** (solve with the right side set to 0) and **P.I.** (deal with the right side). $y = \text{C.F.} + \text{P.I.}$

11 C.F. + P.I. — solve $(D^2 + 2D + 1)y = x e^x$

6x (topic)

light calculus

Step 1 — the C.F. Replace D by m and set to zero:

$$m^2 + 2m + 1 = 0 \rightarrow (m+1)^2 = 0 \rightarrow m = -1, \text{ twice}$$

$$\text{Repeated root rule: } y^c = (C_1 + C_2x)e^{-x}$$

$$\text{Step 2 — the P.I. } y_p = [1/(D^2+2D+1)] \cdot x e^x$$

The right side is $e^a x$ times something, with $a = 2$. Use the **shift rule**: pull the exponential out front and **replace every D by $D+a$** :

$$y_p = e^2 x \cdot [1/((D+2)^2 + 2(D+2) + 1)] \cdot x$$

Step 3 — expand the bracket.

$$(D+2)^2 = D^2 + 4D + 4; \quad 2(D+2) = 2D + 4$$

$$\text{Add them and the 1: } D^2 + 4D + 4 + 2D + 4 + 1 = D^2 + 6D + 9$$

$$y_p = e^2 x \cdot [1/(9 + 6D + D^2)] \cdot x$$

Step 4 — expand the fraction as a series. Take the 9 out front so it starts with 1:

$$1/(9 + 6D + D^2) = (1/9) \cdot 1/[1 + (6D + D^2)/9]$$

$$\text{Now use } 1/(1+u) = 1 - u + u^2 - \dots \text{ with } u = (6D+D^2)/9:$$

$$= (1/9)[1 - 6D/9 - D^2/9 + \dots]$$

Why we can stop: we are applying this to x , and $D^2(x) = 0$ (differentiate x twice and you get 0). So only the 1 and the D terms survive.

Step 5 — apply it to x . D means differentiate: $D(x) = 1$.

$$(1/9)[x - (6/9)(1)] = (1/9)[x - 2/3] = (3x - 2)/27$$

$$\text{Answer } y = (C_1 + C_2x)e^{-x} + (3x - 2)e^2x/27$$

The four P.I. rules, in one place

- Right side $e^a x$ → answer is $e^a x / f(a)$: just put $D = a$. If that gives 0, multiply by x and differentiate the bottom, then try again.
- Right side $\sin ax$ or $\cos ax$ → replace every D^2 by $-a^2$.
- Right side x^n → expand the fraction in **rising** powers of D and stop once D^{n+1} would kill it.
- Right side $e^a x \cdot V$ → pull $e^a x$ out front, replace D by $D+a$.

12 Cauchy's equation: $x^2 y'' - 4xy' + 6y = \cos(2 \log x)$

4x

no calculus after the swap

How to spot it The powers of x match the number of dashes: x^2 with y'' , x with y' .

Step 1 — the substitution. Put $x = e^z$, i.e. $z = \log x$. Now D means d/dz . Learn these three replacements:

$$x y' = Dy \quad x^2 y'' = D(D-1)y \quad x^3 y''' = D(D-1)(D-2)y$$

The $(D-1)$ is what people forget. It is *not* D^2 .

Step 2 — substitute.

$$D(D-1)y - 4Dy + 6y = \cos 2z \quad (\log x \text{ became } z)$$

Expand $D(D-1) = D^2 - D$:

$$D^2 y - Dy - 4Dy + 6y = \cos 2z \rightarrow (D^2 - 5D + 6)y = \cos 2z$$

It is now an ordinary constant-coefficient equation.

Step 3 — C.F. $m^2 - 5m + 6 = 0 \rightarrow (m-2)(m-3) = 0 \rightarrow m = 2, 3$

$$y^c = C_1 e^{2z} + C_2 e^{3z}$$

Step 4 — P.I. $y_p = [1/(D^2 - 5D + 6)] \cos 2z$.

Right side is a cosine $\rightarrow$ **replace D^2 by $-a^2 = -4$** :

$$= [1/(-4 - 5D + 6)] \cos 2z = [1/(2 - 5D)] \cos 2z$$

Step 5 — clear the D from the bottom. Multiply top and bottom by $(2 + 5D)$:

$$= (2 + 5D) \cos 2z / [(2-5D)(2+5D)] = (2 + 5D) \cos 2z / (4 - 25D^2)$$

On the bottom, replace D^2 by -4 again: $4 - 25(-4) = 4 + 100 = 104$

Step 6 — apply the top. $D(\cos 2z) = -2 \sin 2z$:

$$(2 + 5D) \cos 2z = 2 \cos 2z + 5(-2 \sin 2z) = 2 \cos 2z - 10 \sin 2z$$

$$y_p = (2 \cos 2z - 10 \sin 2z) / 104 = (\cos 2z - 5 \sin 2z) / 52$$

Step 7 — go back to x . $e^z = x$, so $e^{2z} = x^2$ and $e^{3z} = x^3$; and $z = \log x$.

Answer $y = C_1 x^2 + C_2 x^3 + [\cos(2 \log x) - 5 \sin(2 \log x)] / 52$

13 Legendre's equation: $(2x+3)^2 y'' + 2(2x+3)y' - 12y = 0$

4x

no calculus after the swap

Same as Cauchy, one extra factor. Put $2x+3 = e^z$. Here $a = 2$ (whatever multiplies x). The replacements pick up powers of a :

$$(2x+3)y' = 2Dy \quad (2x+3)^2 y'' = 2^2 D(D-1)y = 4D(D-1)y$$

The 4 is the mark-loser. Do not drop it.

Step 1 — substitute.

$$4D(D-1)y + 2 \cdot (2Dy) - 12y = 0$$

$$4(D^2 - D)y + 4Dy - 12y = 0$$

$$4D^2 y - 4Dy + 4Dy - 12y = 0$$

The $-4Dy$ and $+4Dy$ cancel: $4D^2 y - 12y = 0$

Step 2 Divide by 4: $D^2 y - 3y = 0$. A.E.: $m^2 - 3 = 0 \rightarrow m = \pm\sqrt{3}$

Step 3 $y = C_1 e^{\sqrt{3}z} + C_2 e^{-\sqrt{3}z}$

Step 4 — back-substitute. Since $e^z = 2x+3$: $e^{\sqrt{3}z} = (e^z)^{\sqrt{3}} = (2x+3)^{\sqrt{3}}$

Answer $y = C_1 (2x+3)^{\sqrt{3}} + C_2 (2x+3)^{-\sqrt{3}}$

Useful fact Cauchy's equation is just Legendre's with $a=1$, $b=0$. Learn one, you have both.

14 Variation of parameters: $y'' - 6y' + 9y = e^{3x}/x^2$

5x — every paper

NEEDS CALCULUS

When to use it When the right-hand side is something the ordinary P.I. rules cannot handle — $\sec^3 x$, $\tan x$, e^{3x}/x^2 , $\cos(e^{-x})$. That awkwardness is the signal.

The recipe For $y'' + Py' + Qy = R$ with C.F. $C_1y_1 + C_2y_2$:

1. $W = y_1y_2' - y_2y_1'$ (the Wronskian)
2. $A = -\int (y_2R/W)dx$ $B = \int (y_1R/W)dx$ (note the minus on A)
3. $y_p = A y_1 + B y_2$

The coefficient of y'' must be 1 before you read off R. Here it already is.

Step 1 — C.F. $m^2 - 6m + 9 = (m-3)^2 = 0 \rightarrow m = 3$ twice

So $y_1 = e^{3x}$ and $y_2 = x e^{3x}$. Also $R = e^{3x}/x^2$.

Step 2 — the two derivatives.

$$y_1' = 3e^{3x}$$

$$y_2' : \text{this is } x \text{ times } e^{3x} \rightarrow \text{product rule. } u = x, u' = 1; v = e^{3x}, v' = 3e^{3x}$$

$$y_2' = (1)e^{3x} + x(3e^{3x}) = e^{3x}(1 + 3x)$$

Step 3 — the Wronskian.

$$W = e^{3x} \cdot e^{3x}(1+3x) - x e^{3x} \cdot 3e^{3x}$$

$$= e^{6x}(1 + 3x) - 3x e^{6x} = e^{6x}(1 + 3x - 3x) = e^{6x}$$

Nice and clean — the x terms cancel.

Step 4 — the first integral.

$$y_2R/W = [x e^{3x} \times e^{3x}/x^2] / e^{6x}$$

Top: $x e^{3x} \times e^{3x}/x^2 = e^{6x}/x$ (the x cancels one x^2)

Divide by e^{6x} : $= 1/x$

$$A = -\int (1/x)dx = -\log x$$

Step 5 — the second integral.

$$y_1R/W = [e^{3x} \times e^{3x}/x^2]/e^{6x} = (e^{6x}/x^2)/e^{6x} = 1/x^2$$

$$B = \int x^{-2}dx = x^{-1}/(-1) = -1/x$$

Step 6 — assemble.

$$y_p = A y_1 + B y_2 = (-\log x)(e^{3x}) + (-1/x)(x e^{3x})$$

The second term: $(-1/x)(x e^{3x}) = -e^{3x}$

$$y_p = -e^{3x} \log x - e^{3x}$$

Step 7 Add the C.F. The lone $-e^{3x}$ is absorbed into C_1 .

Answer $y = (C_1 + C_2x - \log x)e^{3x}$

Even if the integrals defeat you — write the C.F., the Wronskian, and the two integral expressions. That is most of the 7 marks before you integrate anything.

15 Newton's divided difference

5x — every paper

NO CALCULUS

Question Find the cubic through (2,4), (4,56), (9,711), (10,980) and estimate y at $x = 1.5$.

What a "divided difference" is An ordinary difference is just (next – this). A *divided* difference also divides by the gap in x — because the x -values are not evenly spaced.

Step 1 — first divided differences. (y difference) ÷ (x difference) :

$$[2,4] = (56 - 4)/(4 - 2) = 52/2 = 26$$

$$[4,9] = (711 - 56)/(9 - 4) = 655/5 = 131$$

$$[9,10] = (980 - 711)/(10 - 9) = 269/1 = 269$$

Step 2 — second divided differences. Difference of the ones above, divided by the x -gap **spanning both** (first x to last x , skipping one):

$$[2,4,9] = (131 - 26)/(9 - 2) = 105/7 = 15 \quad \leftarrow \text{bottom is } 9-2, \text{ not } 9-4$$

$$[4,9,10] = (269 - 131)/(10 - 4) = 138/6 = 23$$

Step 3 — third divided difference.

$$[2,4,9,10] = (23 - 15)/(10 - 2) = 8/8 = 1$$

X	Y	1ST	2ND	3RD
2	4	26	15	1
4	56	131	23	
9	711	269		
10	980			

Step 4 — build the polynomial. Take the **top diagonal**: 4, 26, 15, 1.

$$P(x) = 4 + 26(x-2) + 15(x-2)(x-4) + 1 \cdot (x-2)(x-4)(x-9)$$

Each new term multiplies in one more bracket, using the x -values in order.

Step 5 — multiply it out.

$$26(x-2) = 26x - 52$$

$$15(x-2)(x-4) = 15(x^2 - 6x + 8) = 15x^2 - 90x + 120$$

$$(x-2)(x-4) = x^2 - 6x + 8; \text{ times } (x-9) : x^3 - 6x^2 + 8x - 9x^2 + 54x - 72 = x^3 - 15x^2 + 62x - 72$$

Add everything: $4 + 26x - 52 + 15x^2 - 90x + 120 + x^3 - 15x^2 + 62x - 72$

$$x^3 : x^3 \cdot x^2 : 15x^2 - 15x^2 = 0 \cdot x : 26 - 90 + 62 = -2 \cdot \text{numbers: } 4 - 52 + 120 - 72 = 0$$

Answer **$P(x) = x^3 - 2x$** $P(1.5) = 3.375 - 3 = \mathbf{0.375}$

If you get stray x^2 or constant terms, you slipped — it really is this clean.

16 Grouped-frequency estimation 3x NO CALCULUS

Question Estimate the number of students scoring between 70 and 75.

MARKS	30-40	40-50	50-60	60-70	70-80
STUDENTS	31	42	51	35	31

Step 1 — THE WHOLE TRICK. You cannot interpolate the counts directly. Convert to "**how many scored less than**" by adding up as you go:

less than 40 → 31

less than 50 → $31 + 42 = 73$

less than 60 → $73 + 51 = 124$

less than 70 → $124 + 35 = 159$

less than 80 → $159 + 31 = 190$

Step 2 — difference table on those cumulative values (ordinary differences now, $h = 10$):

X	Y	Δ	Δ^2	Δ^3	Δ^4
40	31	42	9	-25	37
50	73	51	-16	12	
60	124	35	-4		
70	159	31			
80	190				

Step 3 — find p. $p = (x - x_0)/h = (75 - 40)/10 = 3.5$

Step 4 — Newton forward formula.

$$y = y_0 + p\Delta + [p(p-1)/2]\Delta^2 + [p(p-1)(p-2)/6]\Delta^3 + [p(p-1)(p-2)(p-3)/24]\Delta^4$$

Work out each coefficient with $p = 3.5$:

$$p\Delta = 3.5 \times 42 = 147.0000$$

$$p(p-1)/2 = (3.5)(2.5)/2 = 4.375; \text{ times } \Delta^2 = 9 \rightarrow 39.3750$$

$$p(p-1)(p-2)/6 = (3.5)(2.5)(1.5)/6 = 2.1875; \text{ times } \Delta^3 = -25 \rightarrow -54.6875$$

$$p(p-1)(p-2)(p-3)/24 = (3.5)(2.5)(1.5)(0.5)/24 = 0.2734; \text{ times } \Delta^4 = 37 \rightarrow 10.1172$$

Step 5 — add up.

$$y(75) = 31 + 147.0000 + 39.3750 - 54.6875 + 10.1172 = 172.8047$$

So about **172.80** students scored *less than* 75.

Step 6 — SUBTRACT. You were asked for 70 to 75, and 159 scored less than 70:

$$172.80 - 159 = 13.80$$

Answer **About 14 students**

17 Simpson's rules and the trapezoidal rule 5x NO CALCULUS

Question Evaluate $\int_0^1 dx/(1+x^2)$ using Simpson's 1/3 rule with seven ordinates, and hence find π .

Step 1 — ordinates vs intervals. "Seven ordinates" means seven *y-values*, so $n = 7 - 1 = 6$ intervals. Then $h = (1-0)/6 = 1/6 = 0.166667$.

Step 2 — the y-values. $y = 1/(1+x^2)$ at each x :

I	0	1	2	3	4	5	6
X	0	0.16667	0.33333	0.5	0.66667	0.83333	1
Y	1.000000	0.972973	0.900000	0.800000	0.692308	0.590164	0.500000

e.g. at $x = 1/6$: $x^2 = 1/36$, $1+x^2 = 37/36$, $y = 36/37 = 0.972973$

Step 3 — the rule. $\int y \, dx = (h/3)[(\text{first} + \text{last}) + 4(\text{odd-numbered}) + 2(\text{even-numbered, inside})]$

Odd positions are y_1, y_3, y_5 . Even inside positions are y_2, y_4 .

Step 4 — substitute.

first + last = $1.000000 + 0.500000 = 1.500000$

odds: $0.972973 + 0.800000 + 0.590164 = 2.363137$; times 4 = 9.452548

evens: $0.900000 + 0.692308 = 1.592308$; times 2 = 3.184616

Sum = $1.500000 + 9.452548 + 3.184616 = 14.137164$

$I = (0.166667/3) \times 14.137164 = 0.0555556 \times 14.137164$

Answer **$I = 0.785398$**

Step 5 — get π . The exact value of this integral is $\pi/4$. So

$\pi = 4 \times 0.785398 = \mathbf{3.141592}$ (true value 3.141593 — six decimals correct)

The other two rules on the same numbers

Trapezoidal = $(h/2)[(\text{first}+\text{last}) + 2(\text{everything between})]$. With 4 strips: **0.782794**

Simpson's 3/8 = $(3h/8)[(\text{first}+\text{last}) + 3(\text{all not divisible by 3}) + 2(\text{every third})]$. With 6 intervals: **0.785396**

Which one when: Simpson's 1/3 needs n **even**; Simpson's 3/8 needs n a **multiple of 3**; trapezoidal works for any n . $n=6$ satisfies both Simpsons — which is why papers love it.

18 Lagrange's interpolation 3x NO CALCULUS

Question Find $f(1)$ from $(-1, 9)$, $(0, 5)$, $(2, 3)$.

The pattern One term per data point. Each term is

$y_i \times [(x \text{ minus the other } x\text{'s})] / [(\text{this } x \text{ minus the other } x\text{'s})]$

Step 1 — term for $(-1, 9)$. The other x 's are 0 and 2:

top = $(x-0)(x-2)$ bottom = $(-1-0)(-1-2) = (-1)(-3) = 3$

Term = $9(x)(x-2)/3$

Step 2 — term for $(0, 5)$. Others are -1 and 2:

top = $(x+1)(x-2)$ bottom = $(0+1)(0-2) = (1)(-2) = -2$

Term = $5(x+1)(x-2)/(-2)$

Step 3 — term for $(2, 3)$. Others are -1 and 0:

top = $(x+1)(x-0)$ bottom = $(2+1)(2-0) = (3)(2) = 6$

Term = $3(x+1)(x)/6$

Step 4 — put $x = 1$ straight in (no need to expand):

Term 1: $9(1)(1-2)/3 = 9(1)(-1)/3 = -3$

Term 2: $5(2)(-1)/(-2) = -10/-2 = 5$

Term 3: $3(2)(1)/6 = 6/6 = 1$

Add: $-3 + 5 + 1$

Answer **$f(1) = 3$** (the full polynomial, if asked, is $x^2 - 3x + 5$)

Inverse version If they give you y and want x , use the **identical formula with x and y swapped everywhere**. Nothing else changes.

19 System of ODEs by the matrix method **3x/3 — the last question of every paper** **NO CALCULUS**

Question Rabbits r and wolves w : $dr/dt = 4r - 2w$, $dw/dt = r + w$, with $r(0)=240$, $w(0)=300$.

Step 1 — write it as a matrix. Read the coefficients straight off:

$$A = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}$$

Step 2 — eigenvalues. Subtract λ down the diagonal and set the determinant to zero:

$$\begin{vmatrix} 4-\lambda & -2 \\ 1 & 1-\lambda \end{vmatrix} = (4-\lambda)(1-\lambda) - (-2)(1) = 0$$

$$(4-\lambda)(1-\lambda) = 4 - 4\lambda - \lambda + \lambda^2 = \lambda^2 - 5\lambda + 4$$

$$\text{So } \lambda^2 - 5\lambda + 4 + 2 = 0 \rightarrow \lambda^2 - 5\lambda + 6 = 0 \rightarrow (\lambda-2)(\lambda-3) = 0$$

$$\lambda = 2 \text{ and } \lambda = 3$$

Check: they should add to the trace $4+1 = 5$ ✓ and multiply to the determinant 6 ✓

Step 3 — eigenvector for $\lambda = 2$. Solve $(A - 2I)X = 0$:

$$\begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \rightarrow \text{first row: } 2x_1 - 2x_2 = 0 \rightarrow x_1 = x_2$$

Pick the simplest: $V_1 = [1, 1]^T$

Step 4 — eigenvector for $\lambda = 3$. $(A - 3I)X = 0$:

$$\begin{bmatrix} 1 & -2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \rightarrow x_1 = 2x_2 \rightarrow V_2 = [2, 1]^T$$

Step 5 — write the general solution. One term per eigenvalue:

$$\begin{bmatrix} r \\ w \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{3t}$$

Reading off each row:

$$r = c_1 e^{2t} + 2c_2 e^{3t} \quad w = c_1 e^{2t} + c_2 e^{3t}$$

Keep the eigenvector components with the right variable — the 2 belongs to r .

Step 6 — use the starting values. At $t = 0$, $e^0 = 1$:

$$c_1 + 2c_2 = 240 \quad \dots(i) \quad c_1 + c_2 = 300 \quad \dots(ii)$$

Subtract (ii) from (i): $c_2 = -60$. Then from (ii): $c_1 = 300 + 60 = 360$

Answer $r(t) = 360e^{2t} - 120e^{3t}$ $w(t) = 360e^{2t} - 60e^{3t}$

Check (always do this) $r(0) = 360 - 120 = 240$ ✓ $w(0) = 360 - 60 = 300$ ✓

20 Diagonalization, and hence A^n **3x/3** **NO CALCULUS**

Question Diagonalize $A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$ and hence find A^8 .

Why this works A^8 by hand is eight matrix multiplications. But if $A = PDP^{-1}$ with D diagonal, then $A^8 = PD^8P^{-1}$ — and raising a *diagonal* matrix to a power just means raising each diagonal entry.

Step 1 — eigenvalues.

$$\begin{vmatrix} 1-\lambda & -1 \\ 2 & 4-\lambda \end{vmatrix} = (1-\lambda)(4-\lambda) - (-1)(2) = 0$$

$$(1-\lambda)(4-\lambda) = 4 - \lambda - 4\lambda + \lambda^2 = \lambda^2 - 5\lambda + 4$$

$$\lambda^2 - 5\lambda + 4 + 2 = 0 \rightarrow (\lambda-2)(\lambda-3) = 0 \rightarrow \lambda = 2, 3$$

Step 2 — eigenvectors.

$$\lambda=2: (A-2I) = \begin{bmatrix} -1 & -1 \\ 2 & 2 \end{bmatrix}; \text{row 1 gives } -x_1 - x_2 = 0 \rightarrow x_2 = -x_1 \rightarrow V_1 = [-1, 1]^T$$

$$\lambda=3: (A-3I) = \begin{bmatrix} -2 & -1 \\ 2 & 1 \end{bmatrix}; \text{row 1 gives } -2x_1 - x_2 = 0 \rightarrow x_2 = -2x_1 \rightarrow V_2 = [-2, 1]^T$$

Step 3 — build P and D . Eigenvectors as *columns*, eigenvalues on the diagonal **in the same order**:

$$P = \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Step 4 — invert P . For a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$: swap a and d , negate b and c , divide by the determinant.

$$\det P = (-1)(2) - (-1)(1) = -2 + 1 = -1$$

$$P^{-1} = (1/-1) \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} -2 & -1 \\ 1 & 1 \end{bmatrix}$$

Step 5 — raise D to the 8th. Diagonal, so just do each entry:

$$2^8 = 256, \quad 3^8 = 6561 \rightarrow D^8 = \begin{bmatrix} 256 & 0 \\ 0 & 6561 \end{bmatrix}$$

Step 6 — multiply out $P D^8 P^{-1}$.

$$\text{First } P D^8 = \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 256 & 0 \\ 0 & 6561 \end{bmatrix} = \begin{bmatrix} -256 & -6561 \\ 256 & 13122 \end{bmatrix}$$

Then times $P^{-1} = \begin{bmatrix} -2 & -1 \\ 1 & 1 \end{bmatrix}$:

$$\text{row 1: } (-256)(-2) + (-6561)(1) = 512 - 6561 = -6049; \quad (-256)(-1) + (-6561)(1) = 256 - 6561 = -6305$$

$$\text{row 2: } (256)(-2) + (13122)(1) = -512 + 13122 = 12610; \quad (256)(-1) + (13122)(1) = -256 + 13122 = 12866$$

Answer $A^8 = \begin{bmatrix} -6049 & -6305 \\ 12610 & 12866 \end{bmatrix}$

21 Gauss elimination — the dietician problem 2× NO CALCULUS

Question How much of each food gives 200 mg vitamin A, 250 mg vitamin C, 300 mg calcium?

	FOOD 1	FOOD 2	FOOD 3	NEEDED
Vitamin A	10	30	20	200
Vitamin C	50	30	25	250
Calcium	60	20	40	300

Step 1 — write the equations. Read the table **by row**: one nutrient = one equation. Let x , y , z be grams of Foods 1, 2, 3.

$$10x + 30y + 20z = 200 \quad \dots(1)$$

$$50x + 30y + 25z = 250 \quad \dots(2)$$

$$60x + 20y + 40z = 300 \quad \dots(3)$$

Step 2 — simplify first. Divide (1) by 10, (2) by 5, (3) by 20:

$$x + 3y + 2z = 20 \quad \dots(1')$$

$$10x + 6y + 5z = 50 \quad \dots(2')$$

$$3x + y + 2z = 15 \quad \dots(3')$$

Step 3 — eliminate x .

$$(2') - 10 \times (1'): 6y - 30y = -24y; 5z - 20z = -15z; 50 - 200 = -150$$

$$\rightarrow -24y - 15z = -150, \text{ divide by } -3: 8y + 5z = 50 \quad \dots(4)$$

$$(3') - 3 \times (1'): y - 9y = -8y; 2z - 6z = -4z; 15 - 60 = -45$$

$$\rightarrow -8y - 4z = -45, \text{ i.e. } 8y + 4z = 45 \quad \dots(5)$$

Step 4 — eliminate y . (4) - (5): $5z - 4z = 50 - 45 \rightarrow z = 5$

Step 5 — back-substitute.

$$\text{Into (4): } 8y + 25 = 50 \rightarrow 8y = 25 \rightarrow y = 3.125$$

$$\text{Into (1')}: x + 3(3.125) + 2(5) = 20 \rightarrow x + 9.375 + 10 = 20 \rightarrow x = 0.625$$

Answer **Food 1 = 0.625 g, Food 2 = 3.125 g, Food 3 = 5 g**

Check Vitamin A: $10(0.625) + 30(3.125) + 20(5) = 6.25 + 93.75 + 100 = 200 \quad \checkmark$

22 Rank and consistency 2× NO CALCULUS

Question Find the condition on a , b , c for $x + 2y - z = a$; $2x + y + z = b$; $x - y + 2z = c$ to have a solution.

Step 1 — is the determinant zero?

$$\begin{vmatrix} 1 & 2 & -1 \\ 2 & 1 & 1 \\ 1 & -1 & 2 \end{vmatrix}$$

$$= 1(1 \times 2 - 1 \times (-1)) - 2(2 \times 2 - 1 \times 1) + (-1)(2 \times (-1) - 1 \times 1)$$

$$= 1(2+1) - 2(4-1) - 1(-2-1) = 3 - 6 + 3 = 0$$

Determinant is **zero**, so there is no unique solution — a condition is needed.

Step 2 — find the relation between the rows. Add row 1 and row 3:

$$(1+1)x + (2-1)y + (-1+2)z = a + c \rightarrow 2x + y + z = a + c$$

But row 2 says $2x + y + z = b$.

Step 3 — the two must agree.

Answer **The system is consistent only if $b = a + c$**

If $b \neq a+c$ the same left side would equal two different numbers — impossible, so no solution.

Step 4 — if they ask for the solutions too. Say $a=1$, $b=3$, $c=2$ (so $b=a+c \quad \checkmark$). Rank is 2 with 3 unknowns, so $3-2 = 1$ free parameter. Let $z = t$:

$$\text{From (1): } x + 2y = 1 + t; \text{ from (2): } 2x + y = 3 - t$$

Solving these two: **$x = 5/3 - t$, $y = t - 1/3$, $z = t$** — infinitely many solutions.

Where to spend tonight

Calculus-free 7-markers, worth 14 marks: divided difference · grouped frequency · Simpson · Lagrange · system of ODEs · diagonalization · Gauss elimination · rank/consistency.

Nearly calculus-free: RK4 · Modified Euler · cooling · Cauchy · Legendre — another 5 questions where the only "calculus" is following a substitution rule.

That is **13 of the 22 questions here** with essentially no differentiation or integration. Only the expansions, Lagrange multipliers, orthogonal trajectories, Taylor's method and variation of parameters genuinely need it — and every derivative those need is worked out above.

And on any 7-marker you cannot finish: write the formula, the setup, and the first two steps. That is routinely 3 of 7 marks for ninety seconds. Never leave one blank.

Every 7-mark type from the last-minute prediction sheet, worked from first principles. Every derivative and integral used appears in the toolkit at the top; nothing below it assumes prior calculus. All numerical results computed and verified.  = appearances across the 6 papers.